

UNIVERSITY OF NEW SOUTH WALES  
School of Mathematics and Statistics

MATH3311 Mathematical Computing for Finance

MATH5335 Computational Methods for Finance

Session 2, 2014

Exercises 7 – Nonlinear equations – Solutions

3. The standard normal cumulative probability function  $\Phi : \mathbb{R} \rightarrow [0, 1]$  (Finance texts often use  $N(x)$  instead of  $\Phi(x)$ ) is

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{u^2}{2}} du.$$

The inverse of the standard cumulative normal probability function is  $\Phi^{-1} : [0, 1] \rightarrow \mathbb{R}$  such that

$$\Phi^{-1}(p) = x \quad \text{where} \quad p = \Phi(x).$$

- (a) For a given  $p \in [0, 1]$ , pose the problem of finding  $\Phi^{-1}(p)$  as the solution of a nonlinear equation  $f(x) = 0$ .

**Answer** For a given  $p \in (0, 1)$

$$x = \Phi^{-1}(p) \iff \Phi(x) = p \iff f(x) = \Phi(x) - p = 0.$$

- (b) Calculate the derivatives  $f'(x)$  and  $f''(x)$ .

**Answer**

$$\begin{aligned} f'(x) &= \Phi'(x) = \phi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \\ f''(x) &= \Phi''(x) = \phi'(x) = \frac{-x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} = -x\phi(x). \end{aligned}$$

- (c) Prove that  $x$  is a simple root of  $f$ .

**Answer** A simple root  $x^*$  has  $f(x^*) = 0$  and  $f'(x^*) \neq 0$ . Here  $f'(x) = \phi(x) > 0$  for all  $x \in \mathbb{R}$ , so any root (solution of  $f(x) = 0$ ) must be a simple root.

- (d) If  $\bar{x}$  is an approximate value of the root  $x$ , with error  $|\bar{x} - x| \approx 10^{-6}$ , estimate the error after one iteration of the following methods:

- i. a linearly convergent method with rate  $\beta = 0.9$ ;
- ii. the Secant method
- iii. Newton's method
- iv. Halley's third order rational method

**Answer** The following answers all assume the method is converging to a simple root.

- i. A linearly convergent (order  $\nu = 1$ ) method with constant  $\beta = 0.9 \in (0, 1)$ , has

$$e^{(k+1)} \approx \beta e^{(k)} \implies e^{(k+1)} \approx 0.9 \times e^{(k)} = 0.9 \times 10^{-6}.$$

- ii. The Secant method is super-linearly convergent (order of convergence  $\nu = 1.618 \in (1, 2)$ ), so

$$e^{(k+1)} \approx \left[ e^{(k)} \right]^\nu \implies e^{(k+1)} \approx \left[ e^{(k)} \right]^{1.618} \implies e^{(k+1)} \approx \left[ 10^{-6} \right]^{1.618} = 1.96 \times 10^{-10}.$$

iii. Newton's method is quadratically convergent (order of convergence  $\nu = 2$ ), so

$$e^{(k+1)} \approx \left[ e^{(k)} \right]^\nu \implies e^{(k+1)} \approx \left[ e^{(k)} \right]^2 \implies e^{(k+1)} \approx \left[ 10^{-6} \right]^2 = 10^{-12}.$$

iv. Halley's third order method, even though you do not know how it is defined, by definition has order of convergence  $\nu = 3$ , so

$$e^{(k+1)} \approx \left[ e^{(k)} \right]^\nu \implies e^{(k+1)} \approx \left[ e^{(k)} \right]^3 \implies e^{(k+1)} \approx \left[ 10^{-6} \right]^3 = 10^{-18}.$$

As this is less than the (double precision) relative machine precision  $\epsilon = 2.2 \times 10^{-16}$ , you would only expect  $e^{(k+1)} \approx \epsilon$ .

This question illustrates the significant differences that the order of convergence makes when the iterates are close to a simple root.